

Module 10

Technical Workouts

1. Understand Machine Learning Fundamentals

- a. What is Machine Learning
- b. How ML differs from traditional programming
- c. Types of Machine Learning
- d. Real-world ML applications
- e. ML project lifecycle

2. Understand ML Problem Types

- a. Classification problems
- b. Regression problems
- c. Clustering problems
- d. Recommendation systems
- e. Forecasting problems

3. Understand Data for Machine Learning

- a. Features and labels
- b. Independent and dependent variables
- c. Training datasets
- d. Validation datasets
- e. Testing datasets

4. Understand the ML Workflow

- a. Data collection
- b. Data preparation
- c. Feature selection
- d. Model training
- e. Model evaluation
- f. Model improvement
- g. Understand Training & Prediction

5. Learning from data

- a. Model fitting
- b. Prediction generation
- c. Generalization
- d. Decision boundaries (conceptual)

6. Understand Common ML Challenges

- a. Overfitting
- b. Underfitting
- c. Bias
- d. Variance
- e. Data quality issues

7. Understand Scikit-Learn Ecosystem

- a. Introduction to Scikit-Learn
- b. Dataset loading
- c. Model creation workflow
- d. Training and evaluation pipeline
- e. ML experimentation process